

Technical University of Denmark

Written examination date: May 16, 2024.

Course Title: Computational Data Analysis

Course number: 02582

Aids allowed: All written aids allowed – no internet

Exam Duration: 4 hours

Weighting: 60%

Additional notes:

The examination is a combination of a multiple-choice test and open questions.

For the multiple choice questions, one or more answers are correct for each question.

There are 20 multiple choice questions. The numbering of the individual questions is given as (1), (2), ..., (20). 1 point is given for every correct answer and -1 points are given for a wrong answer. If a question is unanswered 0 points is given.

The number of points corresponding to specific marks or needed to pass the examination is ultimately determined during censoring.

There are two open questions. They are numbered (21) and (22). You can write your answers on separate papers/documents and hand them in. Remember to mark each document with your name, student number, the page number, and out of how many pages. Also mark each answer with the question number the answer belongs to. Answers for each open question is given from 0 up to 10 points.

Question (1)

Which of the following methods are supervised methods?

- A. ☐ Gaussian Mixture Modeling
- B. ☐ Autoencoder
- C. ☐ K-means Clustering
- D. ☐ Tucker
- E. ☐ None of the above

Question (2)

Which of the following methods handle problems with more variables than observations ($p \gg n$) well?

- A. ☐ The Elastic Net Regression
- B. ☐ Principal Component Analysis
- C. ☐ Ordinary Least Squares Regression
- D. ☐ Random Forest Classification
- E. ☐ None of the above

Question (3)

Which of the following statements are true for Random Forest?

- A. ☐ Choosing a small number of randomly selected variables to try in each split leads to a smaller reduction in variance than choosing a larger number of randomly selected variables.
- B. ☐ The Random Forest algorithm cannot be parallelized because a tree in the forest depends on the previous trees.
- C. ☐ Instead of trees, it is possible to use e.g., a KNN model as the individual models in the Random Forest.
- D. ☐ Fully grown trees are more suitable than smaller trees or stumps as the individual models in a Random Forest.
- E. ☐ None of the above

Question (4)

You happen to select a too small λ in your lasso regularization,

$$\arg \min_{\beta} \|y - X\beta\|_2^2 + \lambda \|\beta\|_1$$

How will that affect the estimated model?

- A. ☐ It will have high bias
- B. ☐ It will have high variance
- C. ☐ Not possible to say
- D. ☐ It will have low bias
- E. ☐ None of the above

Question (5)

Which of the following statements are true for regularized regression algorithms?

- A. ☐ Lasso has a closed form solution.
- B. ☐ Lasso is a path algorithm.
- C. ☐ A path algorithm has the advantage of providing solutions for all relevant regularization values.
- D. ☐ The closed form solutions are derived using asymptotic theory ($N \rightarrow \infty$).
- E. ☐ None of the above

Question (6)

Which of the following statements are true for information criteria? (n is the number of observations and p is the number of variables)

- A. ☐ The in-sample error is defined as the error when we sample a new training set at random.
- B. ☐ AIC works equally well for $n \gg p$ and $n \ll p$ situations.
- C. ☐ BIC works for all supervised models in the course, as long as n is high.
- D. ☐ BIC is comparing models according to their posterior odds.
- E. ☐ None of the above

Question (7)

Which of the following statements are true for running cross validation?

- A. ☐ Normalization of variables should be performed before running cross

- validation.
- B. ☐ A double-loop cross validation can assist in both model selection and model assessment.
 - C. ☐ If a subset of observations is dependent, it can be a good idea to keep this subset in the same cross validation fold.
 - D. ☐ Cross validation provides reasonable estimates of the expected prediction error.
 - E. ☐ None of the above

Question (8)

Which of the following statements are true when considering the multiple testing problem?

- A. ☐ Benjamini-Hochberg's algorithm controls the upper bound of the False Discovery Rate.
- B. ☐ The family-wise error rate is the probability of at most α/M false rejections, where M is the number of tests, and α the significance level of the individual tests.
- C. ☐ Bonferroni correction is useful when the number of tests is large but can be too conservative for small number of tests.
- D. ☐ The plug-in estimate of the False Discovery Rate (as defined in Elements of Statistical Learning) is in general NOT a consistent estimate of the False Discovery Rate.
- E. ☐ None of the above

Question (9)

Which of the following statements are true when considering Classification?

- A. ☐ Gaussian Mixture Models uses the k class labels to fit k Gaussian distributions to the data, one to each class.
- B. ☐ Classification and regression trees (CART) explicitly handle categorical variables by one-hot encoding them.
- C. ☐ Random Forest, Sparse Discriminant Analysis, and Regularized Logistic Regression are all suitable models for problems with more variables than observations.
- D. ☐ The Support Vector Machine gives nonlinear boundaries whenever the dual formulation is used for optimization.
- E. ☐ None of the above

Question (10)

Which of the following statements are true when considering Boosting?

- A. ☐ The forward stagewise additive modelling approach works by adaptively updating the weights for all trees in the boosting, as each new tree is added.
- B. ☐ The forward stagewise additive modelling approach works by consecutively adding trees and corresponding weights to the existing trees in the boosting.
- C. ☐ Using AdaBoost.M1 approximates using the exponential loss in a forward stagewise additive modelling.
- D. ☐ The misclassification loss gives higher weight to mis-specified labels than the exponential loss.

- E. ☐ None of the above

Question (11)

Which of the following statements are true for Random Forests and Variable importance?

- A. ☐ The Gini Variable Importance for variable j is an aggregation of the gini index at every split in the forest containing variable j .
- B. ☐ The OOB Variable Importance aggregates the gini index only of trees that do not contain the OOB samples.
- C. ☐ Proximity plots measure the closeness of variables and thereby gives an idea of the grouping of variables, which occur in a Random Forest, similar to what it does in Ridge regression.
- D. ☐ It is a good idea to use deep (large) trees in the Random Forest since the bias of the Random Forest is the same as that of the individual trees.
- E. ☐ None of the above

Question (12)

Which of the following statements about Neural Networks are true?

- A. ☐ The loss of a neural network autoencoder measures the difference between the input and the output of the network.
- B. ☐ Neural networks rarely overfit to training data.
- C. ☐ A feed forward neural network with 10 inputs and 2 hidden layers each with 2 units (nodes) and one output unit has $(10 \times 2 + 2) + (2 \times 2 + 2) + (2 \times 1 + 1) = 31$ parameters/weights that should be estimated.
- D. ☐ A feed forward neural network with 10 inputs and 2 hidden layers each with 2 units (nodes) and one output unit has $10 \times 2 \times 2 \times 1 = 40$ parameters/weights that should be estimated.
- E. ☐ None of the above

Question (13)

When comparing Least Angle Regression with Coordinate Descent for Lasso, which of the following statements are true:

- A. ☐ Both algorithms can be considered path algorithms.
- B. ☐ Both algorithms update the parameter estimate for one variable at a time.
- C. ☐ Both algorithms provide Lasso solutions, without adjusting the algorithms.
- D. ☐ Coordinate Descent makes more assumptions about data than Least Angle Regression.
- E. ☐ None of the above

Question (14)

Which of the following statements about clustering are true?

- A. ☐ The Manhattan distance is used to measure the absolute distance between two points for categorical variables.
- B. ☐ We can use different distance metrics to perform hierarchical clustering.
- C. ☐ We can use AIC or BIC to find the optimal number of clusters in k-means clustering.

- D. ☐ The linkage function decides if we perform agglomerative or divisive hierarchical clustering.
- E. ☐ None of the above.

Question (15)

Figure 1 illustrates 6 two-dimensional simulated datasets and their K-means clustering solution with two clusters.

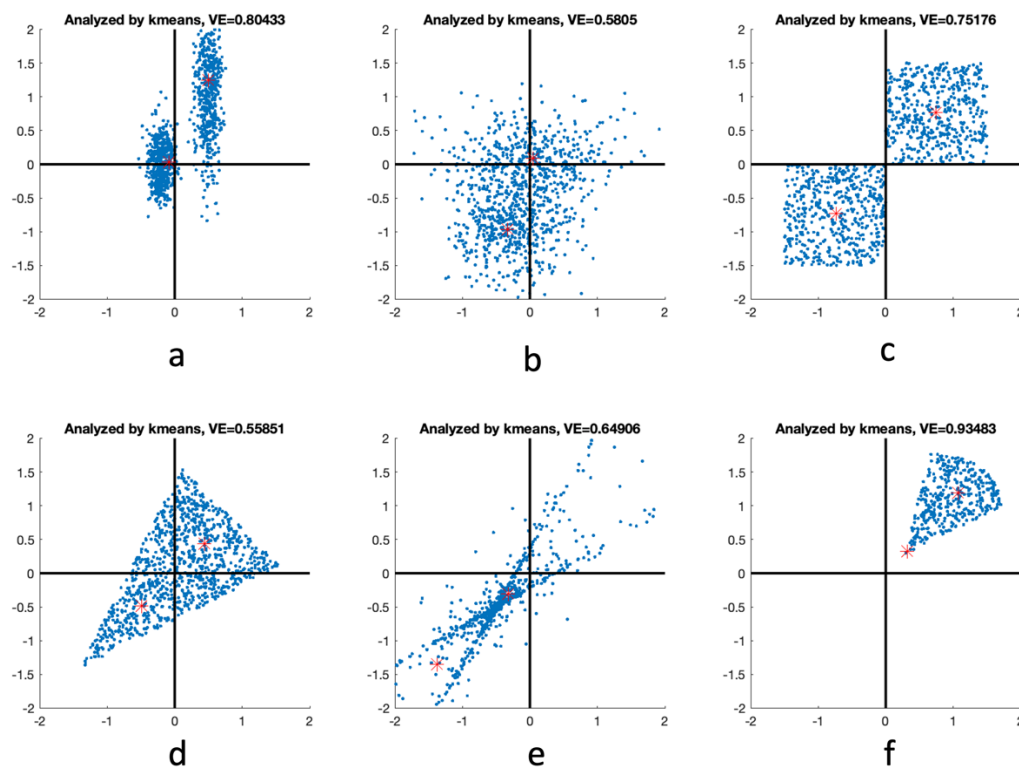


Figure 1

Which of the following statements are true:

- A. ☐ K-means clustering with two clusters describes least of the variance in dataset e than in any of the other datasets.
- B. ☐ K-means clustering with two components is an appropriate choice to describe datasets a and c.
- C. ☐ Archetypal Analysis with three components is an appropriate choice to describe datasets d, e, and f.
- D. ☐ Non-Negative Matrix Factorization (NMF) is an appropriate choice to describe the datasets d and f, without any pre-processing.
- E. ☐ None of the above

Question (16)

For the expected prediction error (EPE), bias and variance of a given model, which of the following statements are true?

- A. ☐ The variance can be estimated as the MSE (Mean squared error) of an independent test set.
- B. ☐ The bias is the difference between the estimate on the training data set and the true value (for the same data).
- C. ☐ The expected prediction error is an expectation over data samples, whereas the generalization error is conditioned on a fixed training set.
- D. ☐ The bias is the difference between the estimate on the test data set and the true value (for the same data).
- E. ☐ None of the above.

Question (17)

Concerning multiway models, which of the following statements are true?

- A. ☐ When the Core Consistency Diagnostic (CONCORDIA) is close to 100, it means that the PARAFAC model has a suitable number of components, because the core tensor is close to diagonal.
- B. ☐ When the Core Consistency Diagnostic (CONCORDIA) is close to 100, it means that the Tucker model has a suitable number of components, because the core tensor is close to diagonal.
- C. ☐ The core tensor in Tucker is a super-diagonal.
- D. ☐ The dimensionality of the core tensor in Tucker defines the ranks in the modes of the decomposition.
- E. ☐ None of the above.

Question (18)

We have data from 1000 features from DNA Expression Microarrays for 820 individuals, all with cancer. We know the cancer type present in each of the 820 individuals and want to investigate if similar DNA Expressions share the same cancer type. Given our aim, which of the following method(s) can help us build a model and answer our question?

- A. ☐ A random forest classification
- B. ☐ A hierarchical clustering
- C. ☐ A random forest regression
- D. ☐ A logistic regression without regularization
- E. ☐ None of the above

Question (19)

Which of the following methods are suitable in terms of achieving a significant reduction in EPE when using the method as the individual models in a bagging procedure?

- A. ☐ Classification and Regression Tree stump.
- B. ☐ K-Nearest Neighbors with a large number of neighbors (K).
- C. ☐ K-Nearest Neighbors with a small number of neighbors (K).
- D. ☐ The mean.
- E. ☐ None of the above

Question (20)

Which of the following methods can be formulated such that the kernel trick can be applied?

- A. ☐ Support Vector Machines
- B. ☐ Boosting
- C. ☐ Principal Component Analysis
- D. ☐ Random Forest
- E. ☐ None of the above

NOTE: The following two questions are open-ended questions.

Question (21):

Demonstrate whether Independent Component Analysis (ICA) is unique or not and describe what it means to have independent components. What kind of distributions are favored for the components of ICA?

Question (22)

You are given a dataset consisting of time series of blood volume pressure (BVP), temperature, and heart rate (HR) measured by a wearable device. These biosignals have been measured for 16 persons under three conditions: During 10 minutes rest, during a 10 minutes running exercise, and during 10 minutes social media usage, and at 4 different time points (one for each season of the year). This results in a total of 192 ($4 \times 16 \times 3$) observations. We are interested in seeing if we can predict the activity of an individual based on the observed biosignals. Consider using a set of extracted biosignal features. Explain which methods you would use to answer whether we can predict the activity or not and to what extend of a) one of the individuals in the experiment, and b) a new individual not previously included in the experiment.